

OPTION THEORY | COLLECTIVE AI | EMPIRICAL FALSIFICATION

FROM SWARM HYPOTHESES TO A GOVERNED AMERICAN OPTION APPROXIMATION

A 50-Agent Research Architecture for Interpretable Early-Exercise Pricing

A pedagogical and research companion to the executable notebook `NB_50_AGENT_AMERICAN_OPTION_CLOSED_FORM_SWARM.ipynb`. The paper studies a hybrid architecture in which large-language-model agents propose economically interpretable structure, while deterministic numerical methods construct the benchmark, estimate coefficients, impose elementary pricing constraints, and adjudicate accuracy.

PAPER PURPOSE

American-style exercise turns an otherwise familiar option-pricing problem into an optimal-stopping problem with a free boundary. This paper asks a different question from the classical derivation literature: can a governed, heterogeneous LLM swarm help discover a compact and interpretable basis for an *approximate* early-exercise premium, provided that the swarm is denied authority over prices, coefficients, benchmarks, and validation? The companion notebook implements this division of labor explicitly. Fifty Claude Sonnet 5 agents, arranged as ten specialist families and five variants, generate structural hypotheses over a controlled feature vocabulary. A separate Cox–Ross–Rubinstein benchmark, ridge-regularized estimation, holdout testing, regime diagnostics, swarm-size convergence analysis, and an extended stress sample determine what survives.

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Abstract

American options embed an exercise-timing decision that makes valuation an optimal-stopping problem rather than a terminal-payoff problem. In the Black–Scholes setting the resulting free boundary generally prevents the existence of a single elementary closed-form solution, motivating lattices, finite-difference methods, simulation-based methods, and a long literature on analytic approximations [3, 13, 6, 1, 2, 5, 11]. This paper develops and evaluates a different research architecture: a heterogeneous swarm of 50 large-language-model agents is used to propose a parsimonious set of dimensionless state variables for an empirical early-exercise-premium approximation, while all numerical truth and statistical adjudication remain outside the swarm.

The companion notebook `NB_50_AGENT_AMERICAN_OPTION_CLOSED_FORM_SWARM.ipynb`¹ [15] generates a synthetic universe of 2,200 option contracts over controlled ranges of moneyness, maturity, volatility, interest rates, and continuous dividend yields. European values are computed analytically and American values are generated with a 300-step Cox–Ross–Rubinstein tree. The data are split 75/25 into a coefficient-estimation sample and a holdout sample. Crucially, the agents do not inspect the raw training rows or holdout outcomes. They receive the research objective, parameter ranges, a governed feature vocabulary, and specialist mandates. Their task is hypothesis generation: identify compact candidate structure, economically relevant interactions, and likely failure modes.

The realized notebook run produced 50 valid structured contracts. Family-level consensus selected 14 features. Separate ridge regressions were then fitted for puts and dividend-paying calls; non-dividend-paying calls were assigned the exact Black–Scholes equality under the maintained assumptions. On the 550-contract holdout set, the final approximation achieved mean absolute error of 0.1047, root mean squared error of 0.1913, median absolute error of 0.0355, and mean absolute error divided by strike of approximately 0.10%. No simple bound violations occurred because the prediction rule explicitly projects onto elementary no-arbitrage bounds. A harder 500-contract stress sample increased MAE to 0.2355 and RMSE to 0.4970, illustrating the expected deterioration outside the central estimation domain. The swarm-size experiment is particularly informative: performance improved materially from 5 to 20 agents, while subsequent expansion to 50 agents added features with little further holdout improvement. The evidence therefore supports a more nuanced claim than “more agents are better”: diversity can add structural information early, but marginal informational value can saturate.

The methodological contribution is the separation of institutional responsibilities. Option theory supplies invariants and decomposition. The agent swarm supplies hypotheses. Numerical methods supply prices and coefficients. Holdout data supplies falsification. Governance supplies provenance and admissibility. The output is not an exact solution of the American-option free-boundary problem. It is an auditable empirical approximation and, more importantly, a reproducible architecture for using collective generative intelligence inside rigorous quantitative-finance research.

CORE EMPIRICAL RESULT

The executed notebook produced a compact, inspectable approximation with holdout MAE = 0.1047, holdout RMSE = 0.1913, and MAE/strike $\approx$ 0.00102, under the synthetic domain defined in the experiment. The stress-domain MAE rose to 0.2355. These numbers should be interpreted as evidence about this specific experimental design, not as a claim of superiority over Barone–Adesi–Whaley, Bjerksund–Stensland, finite differences, or other established methods, which are not directly benchmarked in the current notebook.

¹The Google Colab notebook is available at <https://drive.google.com/file/d/1bxF7kGzfEpTUtHagcFH0JWytaXr5m5eM/view?usp=s haring>.

The American Option Problem and the Research Question

The conceptual difficulty of an American option is easy to state and hard to remove. A European option can be exercised only at maturity. Under the Black–Scholes assumptions, its value can therefore be expressed as the discounted value of a terminal contingent payoff, leading to the celebrated closed-form expressions of Black and Scholes [3] and Merton [13]. An American option adds a right: the holder may exercise at any time before maturity. The value of this right depends on whether immediate exercise dominates the continuation value of waiting.

That additional right transforms valuation into an optimal-stopping problem. At every admissible exercise date the holder compares two economic objects: the value received by exercising now and the value of keeping the option alive. The critical asset price that separates the exercise and continuation regions is not generally known in advance. It is itself part of the solution. In the continuous-time formulation this is the familiar free-boundary problem associated with the American put and related claims [12]. Discrete-time lattices make the same logic transparent: at every node, one takes the maximum of intrinsic value and discounted continuation value [6].

There is an important special case. Under the standard Black–Scholes assumptions, a non-dividend-paying American call is not optimally exercised before maturity; its value therefore coincides with the corresponding European call. The notebook preserves this theorem rather than asking a statistical model to relearn it. Puts and dividend-paying calls, by contrast, may carry a positive early-exercise premium.

The central decomposition in the notebook is therefore

$$V_A(S, K, T, r, q, \sigma) = V_E(S, K, T, r, q, \sigma) + \Pi(S, K, T, r, q, \sigma), \quad (1)$$

where V_E is the corresponding European Black–Scholes price and $\Pi \geq 0$ is the early-exercise premium. This is more than algebraic convenience. It is an economic prior. The experiment does not ask a swarm to rediscover all of option pricing. It asks whether collective generative reasoning can help identify a compact representation of the *incremental* value created by early exercise.

A long literature already provides analytic approximations, bounds, and numerical methods for this problem. Brennan and Schwartz [4], Barone–Adesi and Whaley [1], Bjerk Sund and Stensland [2], Broadie and Detemple [5], Longstaff and Schwartz [11], and Tsitsiklis and Van Roy [16] illustrate distinct approaches to the exercise problem. The present study does not seek to displace that literature. Its research question is methodological:

Can a governed collection of LLM research agents contribute useful *structural hypotheses* to an interpretable American-option approximation, while deterministic numerical methods retain authority over prices, coefficients, admissibility, and validation?

This wording is deliberately restrictive. The swarm is not the pricing engine. It is not the statistical estimator. It is not the final judge. The experiment is designed so that generative intelligence contributes where it is plausibly useful—representation, decomposition, critique, and hypothesis generation— while deterministic computation handles tasks where reproducibility is non-negotiable.

The Classical Benchmark: Why the Swarm Needs an External Judge

A research swarm that proposes an approximation requires an answer key that it does not control. In the notebook, that answer key is numerical rather than linguistic. European call and put prices are computed from Black–Scholes. American prices are generated with a Cox–Ross–Rubinstein binomial tree [6]. The resulting American price is treated as the reference against which the empirical approximation is fitted and evaluated.

The CRR tree is pedagogically attractive because it makes the exercise logic visible. Starting at maturity, option payoffs are known. The tree then works backward. At each node the continuation value is obtained by risk-neutral discounting of the next-period values. The American value at that node is the larger of continuation value and immediate intrinsic value. The free boundary is therefore not imposed by the LLM and is not inferred from a prompt; it emerges from repeated deterministic optimal-stopping comparisons.

The notebook uses 300 time steps for the benchmark. A finite tree is not the same object as the continuous-time mathematical limit, and the paper does not present the tree as exact “ground truth” in an ontological sense. It is better understood as a controlled numerical reference of sufficiently high resolution for the present experiment. A natural robustness extension would compare results against a finer tree and a finite-difference solver.

The most important methodological feature is institutional separation. No agent can alter the benchmark. No agent can manufacture American prices. No agent can decide that its own proposed basis is accurate. This prevents a form of circular validation that would otherwise be easy to overlook in an LLM-heavy workflow.

WHY THIS SEPARATION MATTERS

If the same generative system both proposes a formula and labels its own predictions as correct, the research architecture confounds creativity with verification. Here the benchmark exists before the swarm acts. The agents propose; the numerical layer adjudicates.

The Synthetic Laboratory: What the Dataset Is For

One of the easiest points to misunderstand in the notebook is the role of the synthetic dataset. The data are *not* primarily created to train the 50 LLM agents. The agents do not inspect the 2,200 raw rows and do not observe the holdout outcomes. The dataset has three different users: the coefficient estimator, the holdout evaluator, and the stress/convergence experiments.

The central universe fixes $S = 100$ and samples the other state variables over economically broad but controlled ranges:

State variable	Central experimental range
Log-moneyness $x = \ln(K/S)$	Uniform on $[-0.35, 0.35]$
Maturity T	Approximately 7 days to 3 years, sampled on a log scale
Volatility σ	10% to 60%
Risk-free rate r	0% to 8%
Continuous dividend yield q	0% to 8%
Option type	Random call or put
Spot S	Fixed at 100
Sample size	2,200 contracts

For every contract the notebook stores the European price, the CRR American price, the non-negative difference between the two, and the premium normalized by strike. A fixed random seed then splits the sample into approximately 1,650 training observations and 550 holdout observations.

The training sample is used only *after* the swarm has selected candidate features. Its role is to estimate the coefficients of the premium equation. The holdout sample is withheld from that coefficient-estimation process and serves as an out-of-sample examination. Cell 9 reuses the same train/holdout division for a fair comparison of different swarm sizes. Cell 10 creates an additional 500-contract stress sample over a more extreme parameter region.

WHO USES THE DATA?

- The agents:** receive the research objective, parameter ranges, feature definitions, and specialist mandates, but not the raw outcome rows.
- The ridge estimator:** uses the 75% training sample to estimate coefficients conditional on the swarm-selected basis.
- The validation layer:** uses the 25% holdout sample to measure generalization.
- The stress layer:** generates a new, harder 500-contract sample to probe scope limitations.

This separation gives the experiment a recognizable scientific structure. Economic theory motivates candidate variables. The swarm generates hypotheses about which variables deserve attention. The training data quantify those hypotheses. The holdout data attempt to falsify the resulting equation.

Dimensionless Representation and the Governed Feature Library

The candidate basis is not unconstrained symbolic mathematics. Cell 1 declares a feature vocabulary; Cell 3 computes those features deterministically from the option state. The restriction is important for three reasons.

First, it makes the swarm’s answers comparable. If every agent could invent an arbitrary symbolic representation, aggregation would become ill-defined. Second, it reduces execution risk: agents do not emit executable algebra that could silently change the experiment. Third, dimensionless or economically scaled variables improve portability across nominal price units.

The library includes log-moneyness $x = \ln(K/S)$, absolute and squared moneyness, $\sqrt{T}$, $\sigma\sqrt{T}$, total variance σ^2T , rate-time and dividend-time terms, carry $(r - q)T$, discount factors, moneyness–volatility interactions, rate/dividend

interactions, and hinge-like terms that distinguish call and put in-the-money regions. The final dependent variable is

$$y = \frac{V_A - V_E}{K},$$

(2)

the early-exercise premium normalized by strike. Normalization has two benefits. It removes much of the irrelevant nominal scale and makes the coefficient problem closer to one of describing shape than price level. The design should not be overinterpreted. Dimensionless variables do not guarantee a universal approximation. The feature library is itself a research choice, and all swarm “discovery” occurs conditional on that choice. In statistical language, the agents select within a pre-specified hypothesis space. A different library could lead to a different result even with the same 50 agents.

The 50-Agent Taxonomy: Controlled Heterogeneity Rather Than Randomness

The most distinctive component of the notebook is the construction of fifty research roles from a single underlying LLM. The architecture is 10 specialist families × 5 research lenses. The purpose is not to simulate 50 independent human experts. All agents share the same base model, and their outputs are therefore correlated. The objective is narrower: deliberately induce *structured heterogeneity* in the hypotheses submitted to the downstream numerical layer. The ten families are:

Family	Intellectual role	Question emphasized
Free boundary	Optimal stopping and exercise-region geometry	Which variables proxy distance to the exercise/continuation boundary?
Asymptotics	Limiting regimes	What must the approximation do near expiry or other limiting cases?
Arbitrage bounds	Economic admissibility	Which structures are consistent with monotonicity and basic price bounds?
Dividends and carry	Cash-flow timing	How do r , q , and carry alter early-exercise incentives?
Moneyness geometry	Relative position of spot and strike	How should distance from the strike and the exercise region enter?
Maturity regimes	Horizon dependence	Do short- and long-dated contracts require different scaling?
Volatility structure	Diffusive uncertainty	Which volatility-time and volatility-moneyness interactions matter?
Symbolic regression	Parsimony and interpretability	Which small basis can capture useful nonlinearity without a black box?
Numerical stability	Conditioning and extrapolation	Which terms are stable, non-redundant, and estimable?
Adversarial critic	Falsification	Which omissions are most likely to cause failure?

Each family is instantiated under five lenses: smallest defensible basis, regime interactions, limiting cases, robustness over the stated domain, and a skeptical coauthor mandate. Thus a “volatility agent” is not a unique object; there are five volatility agents with different research attitudes. The same construction is repeated for each family.

What each agent is given

Each agent receives five ingredients:

1. its specialist identity and variant lens;
2. the decomposition of American value into European value plus an early-exercise premium;
3. the central parameter domain;
4. the controlled feature vocabulary; and
5. an explicit output contract and prohibition against fabricating coefficients, prices, benchmark errors, or claims of exact solution.

The agent’s task is therefore much closer to a research committee member than to a pricing engine. It must nominate 3–10 features, optionally emphasize up to four interactions, provide a concise theoretical rationale, name likely failure modes, and state a subjective confidence score.

What “training” does not mean here

No LLM weights are updated. The 50 agents are not trained on the synthetic options. They are prompted using pretrained knowledge and role-specific instructions. The word “training” in later cells refers to statistical coefficient estimation on the 75% sample. This distinction is fundamental: the swarm contributes *prior structural beliefs*; the data estimate their empirical relevance.

Structured Execution, Pre-Flight Control, and Provenance

Cell 5 operationalizes the swarm. It uses a structured JSON schema rather than unconstrained prose. The schema requires family, variant, selected features, must-have interactions, rationale, expected failure modes, and confidence. A one-agent pre-flight call validates authentication, model identity, structured output, and JSON parseability before the remaining 49 agents are launched in parallel.

The implementation also guards against truncated responses. A response that terminates because it reaches the initial output ceiling is retried with a larger budget. Other unexpected stop conditions are recorded as failures. Parallel execution uses a bounded worker pool rather than 50 simultaneous requests.

The realized run reported:

Execution statistic	Observed value
Requested agents	50
Successful structured contracts	50
Failures	0
Input tokens	99,880
Output tokens	24,418
Mean agent time	7.40 seconds
Median agent time	7.02 seconds
Agents requiring retry	0

The governance significance of this cell is easy to understate. A multi-agent experiment is not reproducible simply because all agents use the same model. One must preserve what role was assigned, what the model returned, how long the call took, whether it failed, what token budget was used, and what output was accepted. The notebook therefore treats agent output as provenance-bearing research material rather than ephemeral conversation.

This design is consistent with broader work on role-based and multi-agent LLM systems, where decomposition, standardized operating procedures, memory, tool interfaces, and explicit coordination materially affect system behavior [17, 19, 14, 18, 10]. The present notebook, however, deliberately keeps the agents away from execution tools: their role is hypothesis generation, not autonomous price calculation.

From Fifty Opinions to a Candidate Basis

Raw agent output is not yet evidence. Cell 6 first validates every contract: unknown feature names are discarded, duplicates are collapsed, per-agent complexity is capped, and confidence is clipped to a valid interval. Accepted contracts become rows in a canonical dataframe.

Each selected feature then becomes a vote. The notebook records three summary quantities:

- **agent support:** how many individual agents selected the feature;
- **family support:** how many distinct specialist families selected the feature at least once; and
- **mean confidence:** the average self-reported confidence among agents supporting that feature.

Family support is the central governance correction. Five variants inside one family are correlated by construction. More fundamentally, all 50 agents share one base model. The votes are therefore not independent draws from a well-defined statistical population. Counting specialist families reduces, but does not eliminate, dependence. Consensus should consequently be interpreted as *structured model agreement*, not as a classical sampling probability.

The realized run selected the following 14-feature basis:

Feature	Agent support	Family support	Interpretive role
call_itm	50	10	Call-side hinge moneyness
put_itm	50	10	Put-side hinge moneyness
varT	50	10	Total variance scale
sig_sqrtT	46	10	Diffusive volatility-time scale
carryT	35	10	Net carry over maturity
qT	29	10	Dividend-yield time scale
rT	29	10	Interest-rate time scale
x_rT	33	9	Moneyness–rate interaction
disc_q	25	9	Dividend discount factor
disc_r	16	9	Risk-free discount factor
x_qT	32	8	Moneyness–dividend interaction
sqrtT_q	14	7	Short-horizon dividend scaling
sqrtT_r	13	7	Short-horizon rate scaling
call_itm2	8	6	Nonlinear call-side hinge curvature

The striking result is the breadth of agreement on moneyness, total variance, and volatility-time scaling. Yet unanimity must be read carefully. Shared pretraining, shared wording, and a common permitted vocabulary can all induce correlated selections. The more defensible claim is that, conditional on this prompt architecture and feature library, the swarm exhibited strong and cross-family structural agreement.

Deterministic Estimation: The Dataset Finally Enters the Equation

Only after the feature basis has been selected does the training dataset become active in model construction. Cell 7 fits separate ridge regressions for puts and dividend-paying calls. Let $\phi(S, K, T, r, q, \sigma)$ denote the selected feature vector including an intercept. The normalized premium is modeled as a linear form in that basis:

$$\hat{\pi}_j = \beta_j^\top \phi, \quad j \in \{\text{put}, \text{call}\}, \quad (3)$$

with ridge regularization. The final premium uses a positive-part operation,

$$\hat{\Pi}_j = K \max(0, \hat{\pi}_j), \quad (4)$$

and the reconstructed American value is then projected onto simple bounds.

Ridge regression is a conservative choice. The feature library contains terms that are economically distinct but statistically correlated—for example rT , qT , carry, discount factors, and their interactions. A small L_2 penalty stabilizes coefficient estimation without turning the model into a black box. The notebook uses $\alpha = 10^{-5}$, with the intercept explicitly included in the design matrix. Ridge regression itself has a long statistical history as a response to nonorthogonal predictors [9].

The final pricing forms are

$$\hat{P}_A = \mathcal{B}_P [P_{BS} + K \max(0, \beta_{\text{put}}^\top \phi)], \quad (5)$$

$$\hat{C}_A = \mathcal{B}_C [C_{BS} + K \max(0, \beta_{\text{call}}^\top \phi)], \quad q > 0, \quad (6)$$

where $\mathcal{B}_P$ and $\mathcal{B}_C$ denote deterministic projection onto the notebook’s elementary lower and upper bounds. For a non-dividend-paying call,

$$\hat{C}_A = C_{BS}. \quad (7)$$

The word “closed form” therefore requires qualification. The final approximation is operationally closed-form in the sense that inference requires Black–Scholes, elementary feature transformations, a finite weighted sum, a positive-part operator, and simple bounds. No tree or iterative free-boundary solver is required at inference time. It is *not* an exact closed-form solution of the American free-boundary problem.

Realized coefficients

The executed notebook produced the following coefficient vectors. They are reported because auditability is part of the research claim; they should not be treated as universal constants.

Put premium coefficients		Dividend-call premium coefficients	
Term	Coefficient	Term	Coefficient
Intercept	0.629106	Intercept	0.902217
call_itm	0.000915	call_itm	0.007711
put_itm	0.009212	put_itm	0.006049
varT	-0.001553	varT	0.006429
sig_sqrtT	-0.001024	sig_sqrtT	-0.007588
carryT	0.123673	carryT	-0.043107
qT	-0.277019	qT	-0.325241
rT	-0.153346	rT	-0.368348
x_rT	0.518102	x_rT	0.242281
disc_q	-0.438950	disc_q	-0.433785
disc_r	-0.189996	disc_r	-0.468319
x_qT	-0.281887	x_qT	-0.633514
sqrtT_q	-0.097047	sqrtT_q	0.026813
sqrtT_r	-0.013541	sqrtT_r	-0.107882
call_itm2	0.009509	call_itm2	0.016514

Individual coefficient signs should not be read as isolated structural derivatives because the regressors are correlated and the fitted object is subsequently transformed by a positive-part operator and bound projection. Interpretation is more defensible at the level of the complete basis and predicted premium than at the level of one ceteris-paribus coefficient.

Holdout Validation: Does the Approximation Generalize?

The notebook reserves 550 contracts for holdout validation. These observations do not determine the ridge coefficients. The prediction function generates an American value for each holdout contract and records absolute error, squared error, strike-normalized error, and log-moneyness.

The realized summary metrics are:

Subset	<i>N</i>	MAE	RMSE	Median AE	P95 AE	MAE/ <i>K</i>
All	550	0.104729	0.191275	0.035479	0.401885	0.001017
Put	267	0.114108	0.206522	0.030767	0.440876	0.001042
Call	283	0.095881	0.175680	0.039434	0.350096	0.000994

Because the spot normalization is $S = 100$ and strikes are in the same broad order of magnitude, these errors are economically small within the central synthetic domain. Yet average error is not enough. The notebook therefore evaluates regimes in which the approximation may be more difficult:

Regime	<i>N</i>	MAE	RMSE	Median AE	P95 AE
Short maturity	183	0.059678	0.100360	0.014878	0.214853
High volatility	167	0.086204	0.157616	0.030548	0.310035
Deep ITM or OTM	204	0.134973	0.210715	0.089606	0.437650
Near ATM	80	0.089626	0.199880	0.006101	0.439283
Dividend calls	217	0.113822	0.196537	0.061240	0.390595

The deepest in/out-of-the-money region has a higher MAE than the full sample, consistent with the intuition that a single global linear premium basis will have more difficulty capturing curvature near different exercise regimes. The near-ATM group has a very small median error but a comparatively elevated P95 error, showing why a single average statistic is insufficient.

The notebook also prints the twelve worst holdout cases. The largest absolute error is approximately 1.2064 for a long-dated put. Several of the worst cases have maturities above two years, again suggesting that the hardest observations are not uniformly distributed across the domain.

Arbitrage checks: useful but not evidence of accuracy

The reported counts of prices below intrinsic value, below European value, and above the notebook’s simple upper bounds are all zero. This is desirable but must not be misinterpreted. The prediction function explicitly clips prices to those bounds. Zero violations are therefore a *design invariant*, not an independent empirical achievement. A poor raw regression could still produce zero final bound violations after projection.

This distinction illustrates a general principle in quantitative AI: constraint satisfaction and predictive accuracy are different properties. A model can be economically admissible and statistically poor, or statistically accurate on average while violating critical constraints. A production-grade system requires both.

Does a Larger Swarm Add Information?

A fifty-agent architecture should not be justified by spectacle. Cell 9 treats swarm size as an empirical design variable. Validated contracts are ordered so that early subswarms cover multiple specialist families. Nested groups of 5, 10, 20, 30, 40, and 50 agents are then used to re-select features, refit local put/call models on the same training data, and evaluate them on the same holdout sample.

The realized convergence table is:

Agents	Selected features	Holdout RMSE	Holdout MAE
5	6	0.297652	0.161539
10	6	0.245072	0.153471
20	9	0.187537	0.104987
30	11	0.187997	0.104956
40	13	0.191408	0.105571
50	14	0.191275	0.104729

This result is more interesting than a monotonic improvement story. Moving from 5 to 20 agents produces a substantial reduction in error and a broader basis. Beyond 20 agents, the model continues to acquire features, but holdout RMSE does not improve further and in fact oscillates slightly. MAE is essentially flat from 20 through 50 agents.

The implication is not that twenty is universally optimal. The ordering is one deterministic subswarm construction, the agents share a base model, and the candidate vocabulary is finite. The defensible inference is narrower: *under this architecture and this run, most measurable structural value arrived before all 50 agents were included*. Additional agents increased representational complexity more clearly than predictive accuracy.

That finding matters for agentic-system design. Multi-agent research systems have real costs: tokens, latency, coordination, failure handling, provenance, and review. The optimal number of agents is therefore a problem of marginal information, not prestige.

Extended-Domain Stress Test and the Boundary of the Claim

Cell 10 deliberately leaves the central domain. It generates 500 new contracts with log-moneyness extending to $[-0.55, 0.55]$, maturities from approximately one day to five years, volatility from 8% to 90%, and r and q up to 12%. The same Black–Scholes/CRR reference architecture is applied and the final equation is evaluated without refitting.

The stress results are:

Metric	Stress result
N	500
MAE	0.235526
RMSE	0.497041
Median absolute error	0.109044
95th percentile absolute error	0.814579

The increase in error is not a defect to hide. It is exactly the information a stress test should reveal. The formula was fitted inside a specified region and its accuracy deteriorates as it moves toward more extreme combinations. That behavior reinforces the paper’s preferred interpretation: this is an empirical approximation with a documented domain, not a new theorem of American-option valuation.

The stress test also prevents a common failure in AI-assisted quantitative research: reporting a single attractive holdout number while leaving the domain of validity implicit. Here the domain expansion is explicit and the degradation is measured.

How the Result Sits Within the American-Option Literature

The notebook should be situated carefully relative to established option pricing methods. It does not derive a new free-boundary theorem. Nor does it show that its approximation dominates classic analytic approximations. The current experiment does not compute head-to-head error and speed against Barone–Adesi–Whaley [1], Bjerk Sund–Stensland [2], Broadie–Detemple [5], finite-difference methods, or least-squares Monte Carlo [11].

Its novelty is architectural. Classical methods begin from option theory and derive numerical or analytic devices for approximating the stopping problem. The present workflow inserts a governed generative-research layer between theory and estimation. That layer proposes a sparse representation of the premium. The numerical layer remains conventional and auditable.

Three comparisons are therefore useful.

Compared with analytic approximations

Barone–Adesi–Whaley and Bjerk Sund–Stensland explicitly exploit structural properties of early exercise to derive fast approximations. Their great advantage is theoretical specificity and long-standing validation. The swarm approach is more flexible in representation search but less theoretically committed. It should therefore be judged as a complementary research device, not a substitute for established formulas.

Compared with lattices and PDE methods

Trees and finite differences solve the exercise problem numerically each time a contract is priced. They can be made highly accurate but require repeated backward induction or PDE solution. The fitted swarm-derived approximation moves computation to an offline research phase: once coefficients are fixed, inference is fast. The trade-off is approximation error and domain dependence.

Compared with simulation-regression methods

Longstaff–Schwartz and Tsitsiklis–Van Roy use regression inside dynamic programming or simulation to approximate continuation values [11, 16]. The notebook also uses regression, but for a different object: the final normalized early-exercise premium over a swarm-selected global basis. It is therefore much simpler at inference, but it does not attempt to solve high-dimensional path-dependent stopping problems.

WHAT THE CURRENT EXPERIMENT DOES NOT ESTABLISH

It does not establish universal accuracy, market calibration performance, hedging quality, dominance over existing analytic approximations, robustness to discrete dividends, stochastic volatility, jumps, negative rates, transaction costs, or microstructure effects. Those are research extensions, not hidden assumptions.

Academic Interpretation of the Agent Contribution

The most important empirical object produced by the agents is not a price. It is a *prior over representations*. Each agent maps the same research problem into a small subset of a controlled feature library. The voting stage then converts qualitative theory-conditioned suggestions into an auditable structural prior for the regression.

This interpretation helps bridge LLM language and statistical language.

Agentic-system concept	Statistical interpretation
Specialist family	Structured prior perspective over candidate regressors
Variant lens	Perturbation of the selection criterion
Agent-selected feature	Vote for inclusion in the candidate basis
Family support	Dependence-aware diversity proxy
Rationale	Human-auditable theory statement
Failure mode	Ex ante falsification hypothesis
Confidence	Subjective self-assessment, not a calibrated probability
Consensus basis	Data-independent feature-screening stage
Training regression	Coefficient estimation conditional on screened basis
Holdout evaluation	Out-of-sample adjudication

From this perspective, the research design resembles a two-stage model selection procedure in which the first stage is theory- and LLM-mediated and the second stage is conventional statistical fitting. This is powerful but creates selection considerations. Because the feature library, prompts, and thresholds were designed by the researcher, the final specification is not the output of a purely automatic discovery process. Human design, LLM priors, and numerical evidence all enter.

The notebook’s provenance tables make those layers inspectable, which is a strength. Future versions could make the statistical treatment even stricter by introducing a second untouched test set, repeated random seeds, bootstrap resampling, alternative prompt templates, multiple base models, or nested cross-validation after the swarm screening step.

Limitations and a Research Agenda

The notebook is deliberately transparent about scope. Several extensions are both natural and academically important.

Benchmark the approximation against established formulas

The next empirical paper should run Barone–Adesi–Whaley, Bjerk Sund–Stensland, CRR, and a finite-difference benchmark on the exact same contracts. Error, speed, and stability should be compared by regime. Without that experiment, one should not use words such as “superior” or “state-of-the-art”.

Separate discovery, criticism, and adjudication more strongly

The present swarm includes an adversarial family, but all agents use one base model. A stronger design could use one subswarm to propose features, another independent subswarm or model family to critique them, and a deterministic third layer to design falsification tests. The goal would be to reduce same-model correlated reasoning.

Vary the feature vocabulary

The current result is conditional on a hand-designed library. Future swarms could propose new symbolic primitives, but only through a sandboxed compiler that checks dimensions, numerical stability, and admissibility before a term is eligible for estimation.

Model regimes rather than one global premium surface

The regime diagnostics suggest that one global linear premium basis may be overly restrictive. A small mixture of interpretable regime-specific equations could preserve transparency while allowing different behavior near expiry, deep in/out of the money, and under strong dividend incentives.

Move beyond Black–Scholes dynamics

Discrete dividends, stochastic volatility, jumps, stochastic interest rates, and transaction costs alter the early-exercise problem. Extending the architecture would require both a new numerical benchmark and a revised feature vocabulary. The architecture is portable; the fitted coefficients are not.

Evaluate hedging, not only pricing

A pricing approximation can have small level error and still produce poor sensitivities. A production-oriented extension should study delta, gamma, theta, and exercise-boundary stability. Hedging error may be more relevant than price error for many institutional users.

Treat reproducibility as a first-class quantitative output

The notebook already saves agent provenance, coefficient tables, holdout metrics, and a JSON reproducibility report. Future runs should also archive the exact prompts, model version, API/library version, response schema, hash of the notebook, and figure-generation pipeline. In agentic research, provenance is part of the model specification.

Pedagogical Lessons for Graduate Finance and Financial Professionals

The experiment is useful even for readers who never intend to deploy an LLM swarm in options trading. It makes several deep ideas visible.

First, **an American option is not difficult because Black–Scholes is unknown; it is difficult because the holder owns a timing decision.** The exercise decision creates the free boundary.

Second, **a dataset can play a different role from the one commonly imagined in generative AI.** Here the dataset does not teach the LLM agents. It is the laboratory evidence used after the agents formulate hypotheses.

Third, **multi-agent intelligence is an architectural claim, not a head count.** Fifty identical prompts are not the same experiment as ten specialist families with distinct mandates, controlled outputs, provenance, and family-level consensus.

Fourth, **generative and deterministic intelligence should not be given the same institutional authority.** The agents are good at proposing representations and articulating failure modes. They are not given permission to set benchmark prices or grade their own ideas.

Fifth, **constraints and accuracy are different.** The bound projection guarantees simple admissibility properties, but it does not guarantee that the raw statistical approximation is accurate.

Sixth, **more agents need not mean more information.** The notebook’s own convergence experiment shows diminishing marginal value after the early subswarm gains.

Finally, **an approximate formula is scientifically useful only when its scope is explicit.** The harder stress domain is not an afterthought; it is part of the claim.

THE NOTEBOOK IN ONE SENTENCE

Option theory supplies the invariants; the swarm proposes the representation; training data estimates the coefficients; holdout and stress data try to falsify the result; governance preserves the chain of evidence.

Reproducibility Record

The executed run summarized in this paper corresponds to the companion notebook `NB_50_AGENT_AMERICAN_OPTION_CLOSED_FORM_SWARM.ipynb`. Its recorded configuration is:

Field	Value
Model	claude-sonnet-5
Requested / successful agents	50 / 50
Specialist families	10
Variants per family	5
Random seed	766
CRR steps	300
Ridge penalty	10 ^{−5}
Central sample	2,200 contracts
Training / holdout split	75% / 25%
Holdout observations	550
Stress observations	500
Recorded run timestamp	2026-09-22 14:35:43 UTC

The notebook saves four machine-readable research artifacts: `american_option_swarm_provenance.csv`, `american_option_closed_form_coefficients.csv`, `american_option_holdout_metrics.csv`, and `american_option_reproducibility_report.json`. Together with the notebook itself, these files create a trace from agent proposals to the final equation.

Conclusion

This paper has examined a deliberately hybrid approach to a classic quantitative-finance problem. American options are difficult because early exercise introduces optimal stopping and a free boundary. The notebook does not claim to eliminate that mathematics. Instead, it asks whether a governed collective of language-model agents can contribute to the search for a compact, interpretable empirical representation of the early-exercise premium.

The answer from this experiment is qualified but constructive. The 50-agent swarm produced a coherent, cross-family feature basis. The deterministic estimation layer transformed that basis into separate put and dividend-call premium equations. The resulting approximation generated low errors on the central synthetic holdout set and respected the imposed simple price bounds. Performance deteriorated, as expected, under a deliberately wider stress domain. Most importantly, the convergence experiment showed that the value of additional agents was not monotonic: the strongest accuracy gains appeared before the full 50-agent system was reached.

The research claim is therefore not that fifty LLM agents have solved the American-option free-boundary problem. Nor is it that the resulting formula dominates established analytic approximations. The claim is institutional and methodological. A large generative system can be made useful in serious quantitative research when its responsibilities are sharply bounded.

The swarm is allowed to be imaginative where imagination has value: proposing variables, identifying interactions, expressing theoretical rationale, and anticipating failure. It is denied authority where reproducibility is essential: benchmark construction, coefficient estimation, pricing bounds, holdout evaluation, and stress measurement. The final equation is visible, but the more important artifact is the research architecture that produced it.

This division of labor suggests a broader design principle for AI in finance:

Do not ask a generative model to replace the quantitative institution. Use it as one governed component inside an institution in which theory, computation, data, validation, and accountability remain distinct.

For graduate students, the notebook offers a bridge between derivative-pricing theory and advanced AI systems. For financial professionals, it offers a prototype of how agentic research might be introduced without surrendering auditability. For researchers, it leaves a clear agenda: benchmark against established approximations, repeat across models and seeds, expand the feature space carefully, test hedging behavior, and determine where collective generative intelligence genuinely adds information rather than simply adding computation.

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COMPANION COMPUTATIONAL ARTIFACT

The executable research record for this paper is NB_50_AGENT_AMERICAN_OPTION_CLOSED_FORM_SWARM.ipynb. It contains the pricing benchmark, synthetic universe, governed feature vocabulary, 50-agent mandates, structured execution logic, provenance, family-level consensus, fitted coefficients, holdout diagnostics, swarm-convergence experiment, stress test, and saved machine-readable outputs.